

Analytical feedback and risk evaluation for option hedging under entropy-regularized stochastic control

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Abstract—This paper studies option hedging with market frictions through a stochastic-control formulation. Starting from the Black-Scholes benchmark, I first formulate a deterministic Hamilton-Jacobi-Bellman (HJB) hedging problem with a quadratic position penalty. I then introduce entropy regularization by replacing pointwise deterministic positions with probability distributions over admissible positions. The resulting entropy-regularized HJB equation yields a closed-form local optimizer. On an unconstrained action space, this optimizer is Gaussian. Its mean is the deterministic HJB feedback control, and its variance is governed by the entropy temperature. Under the risk-neutral specification used in the numerical section, the deterministic feedback reduces to a state-dependent shrinkage of the Black-Scholes delta. Based on 100,000 Monte Carlo paths, the numerical study compares the Black-Scholes delta hedge, the analytical deterministic HJB hedge, and the entropy-regularized Gaussian HJB hedge by seller-side hedging P&L, MSE, Value-at-Risk, Conditional Tail Expectation, running position exposure, and the penalized objective. The results show how position shrinkage and entropy-driven randomization change the distribution and tail risk of terminal hedging outcomes.

Index Terms—Stochastic control, Hamilton-Jacobi-Bellman equation, entropy regularization, option hedging, delta hedging, trading frictions.

I. INTRODUCTION

The classical Black-Scholes framework assumes frictionless continuous trading [1], [2]. In that idealized setting, a European option can be replicated by continuously rebalancing the delta position in the underlying asset. Actual markets have transaction costs, liquidity limits, and market impact. Hedging with frictions therefore involves a tradeoff between accurate terminal replication and control of trading intensity.

A standard way to model this tradeoff is to formulate hedging as a stochastic-control problem with a trading or position penalty [3], [4], [6]. Dynamic programming leads to a Hamilton-Jacobi-Bellman equation, and the optimizer is a deterministic feedback strategy [5]. This gives a clear mathematical baseline, but the control remains pointwise. Under model uncertainty, parameter perturbations, or execution noise, a purely deterministic trading rule can be too rigid.

Recent data-driven hedging methods use deep learning and reinforcement learning to handle market frictions, nonlinear instruments, and high-dimensional state variables. The Deep Hedging framework of Buehler, Gonon, Teichmann, and Wood [10], together with later reinforcement-learning implementations such as Cao, Chen, Hull, and Poulos [11], shows that neural-network policies can learn hedging strategies under general frictions. Those methods are flexible, but they often require expensive training and the resulting policies are less

transparent than closed-form feedback rules. The approach here is complementary. I focus on an analytically tractable entropy-regularized HJB structure whose shrinkage factor, exploration variance, and induced risk metrics can be read directly from the formulas.

Motivated by continuous-time exploratory stochastic control [9], this paper studies an entropy-regularized hedging formulation. At each time, the controller chooses a probability distribution over possible positions. The entropy term penalizes overly concentrated policies and introduces a tunable degree of exploration. In the hedging problem, this describes randomized trading while keeping the HJB structure.

The paper gives a transparent application of entropy-regularized relaxed stochastic control to option hedging. The focus is the link between the Gaussian control distribution generated by entropy regularization and measurable hedging outcomes, including P&L, tail risk, the penalized objective, MSE, VaR, and CTE.

The main points are as follows.

- I formulate a wealth-asset hedging model with a quadratic position penalty and derive both deterministic and entropy-regularized HJB equations.
- I solve the local entropy-regularized control problem in closed form and show that the optimal exploratory control distribution is Gaussian on an unconstrained action space.
- I implement the analytical HJB feedback in Monte Carlo simulation and compare the Black-Scholes delta hedge, the deterministic HJB hedge, and the entropy-regularized Gaussian HJB hedge using seller-side P&L, MSE, Value-at-Risk, Conditional Tail Expectation, running position exposure, average shrinkage, and exploration variance.

The remainder of the paper is organized as follows. Section II reviews the Black-Scholes benchmark. Section III presents the deterministic HJB hedging problem. Section IV develops the entropy-regularized stochastic-control model. Section V discusses the indifference-pricing interpretation. Section VI reports the numerical results.

II. BLACK-SCHOLES BENCHMARK

To isolate the effect of trading frictions and control regularization, I use the classical Black-Scholes delta hedge as the benchmark strategy [1], [2]. This benchmark assumes a frictionless market in which the investor can rebalance according to the theoretical delta at each decision time. The assumption is idealized, but it gives a reference point for comparing deterministic control and entropy-regularized control.

probability density $\pi_t(\delta) \in \mathcal{P}(U)$ on an admissible action space $U \subseteq \mathbb{R}$. At time t , the position is sampled from π_t . The control is therefore a distribution, rather than a scalar. This is the hedging version of the relaxed stochastic-control idea in Wang, Zariphopoulou, and Zhou: randomization represents exploration, while a Dirac mass gives the classical deterministic control. The entropy-regularized value function is

$$V(t, s, x) = \inf_{\pi} \mathbb{E}_{t,s,x} \left[(X_T - H(S_T))^2 + \int_t^T \int_U (\lambda \delta^2 + \eta \ln \pi_{\tau}(\delta)) \pi_{\tau}(\delta) d\delta d\tau \right], \quad (6)$$

where $\eta > 0$ is the temperature parameter. In the notation of [9], it is the exploration weight. I reserve λ for the position penalty and use η for the entropy temperature. The term

$$\eta \int_U \pi(\delta) \ln \pi(\delta) d\delta$$

is the negative-entropy regularizer. The terminal condition is

$$V(T, s, x) = (x - H(s))^2. \quad (7)$$

For a smooth test function $V = V(t, s, x)$, the infinitesimal generator under a fixed action δ is

$$\mathcal{L}^{\delta} V = V_t + \mu s V_s + (rx + \delta(\mu - r)s) V_x + \frac{1}{2} \sigma^2 s^2 V_{ss} + \delta \sigma^2 s^2 V_{sx} + \frac{1}{2} \delta^2 \sigma^2 s^2 V_{xx}. \quad (8)$$

Under the exploratory control distribution π , the generator is averaged with respect to $\pi(\delta)$. The entropy-regularized HJB equation is

$$0 = \inf_{\pi \in \mathcal{P}(U)} \int_U \left[V_t + \mu s V_s + (rx + \delta(\mu - r)s) V_x + \frac{1}{2} \sigma^2 s^2 V_{ss} + \delta \sigma^2 s^2 V_{sx} + \frac{1}{2} \delta^2 \sigma^2 s^2 V_{xx} + \lambda \delta^2 + \eta \ln \pi(\delta) \right] \pi(\delta) d\delta. \quad (9)$$

Since

$$\int_U \pi(\delta) d\delta = 1,$$

the terms independent of δ can be moved outside the integral. Hence (9) becomes

$$0 = V_t + \mu s V_s + rx V_x + \frac{1}{2} \sigma^2 s^2 V_{ss} + \inf_{\pi \in \mathcal{P}(U)} \int_U [A \delta^2 + B \delta + \eta \ln \pi(\delta)] \times \pi(\delta) d\delta, \quad (10)$$

where

$$A := \lambda + \frac{1}{2} \sigma^2 s^2 V_{xx}, \quad B := (\mu - r)s V_x + \sigma^2 s^2 V_{sx}. \quad (11)$$

For the minimization problem on $U = \mathbb{R}$ to be well posed, we require

$$A > 0, \quad \text{i.e.} \quad \lambda + \frac{1}{2} \sigma^2 s^2 V_{xx} > 0. \quad (12)$$

The local variational problem is

$$\inf_{\pi \in \mathcal{P}(U)} \int_U [A \delta^2 + B \delta + \eta \ln \pi(\delta)] \pi(\delta) d\delta. \quad (13)$$

Let

$$q(\delta) := A \delta^2 + B \delta. \quad (14)$$

The Lagrangian with normalization constraint $\int_U \pi(\delta) d\delta = 1$ is

$$\mathcal{J}(\pi) = \int_U [q(\delta) + \eta \ln \pi(\delta)] \pi(\delta) d\delta + \lambda_0 \left(\int_U \pi(\delta) d\delta - 1 \right), \quad (15)$$

where λ_0 is the Lagrange multiplier. The first variation in π gives

$$q(\delta) + \eta (\ln \pi^*(\delta) + 1) + \lambda_0 = 0. \quad (16)$$

Thus

$$\pi^*(\delta) = C \exp \left(-\frac{q(\delta)}{\eta} \right), \quad (17)$$

where C normalizes the density. Therefore

$$\pi^*(\delta; t, s, x) = \frac{\exp \left[-\frac{1}{\eta} (A \delta^2 + B \delta) \right]}{\int_U \exp \left[-\frac{1}{\eta} (A \bar{\delta}^2 + B \bar{\delta}) \right] d\bar{\delta}}. \quad (18)$$

If $U = \mathbb{R}$, completing the square gives

$$A \delta^2 + B \delta = A \left(\delta + \frac{B}{2A} \right)^2 - \frac{B^2}{4A}. \quad (19)$$

The optimal exploratory control distribution is Gaussian:

$$\pi^*(\delta; t, s, x) = \mathcal{N} \left(-\frac{B}{2A}, \frac{\eta}{2A} \right). \quad (20)$$

Substituting the definitions of A and B yields

$$\pi^*(\delta; t, s, x) = \mathcal{N} \left(-\frac{(\mu - r)s V_x + \sigma^2 s^2 V_{sx}}{2\lambda + \sigma^2 s^2 V_{xx}}, \frac{\eta}{2\lambda + \sigma^2 s^2 V_{xx}} \right). \quad (21)$$

As $\eta \rightarrow 0$, the variance converges to zero and the exploratory control distribution collapses to the classical deterministic feedback control.

A. Closed-form feedback under the risk-neutral setting

The numerical experiments below use a special case in which the feedback formula can be written directly in terms of the Black-Scholes delta. This keeps the numerical strategy tied to the analytical control, rather than to an additional heuristic rule.

Proposition IV.1 (Risk-neutral analytical feedback). *Assume $\mu = r$, and suppose that the value function has the quadratic form*

$$V(t, s, x) = a(t, s)x^2 + b(t, s)x + c(t, s),$$

with terminal condition $V(T, s, x) = (x - H(s))^2$. Under the Black-Scholes risk-neutral dynamics,

$$a(t, s) = e^{2r(T-t)}, \quad b(t, s) = -2e^{2r(T-t)}C^{BS}(t, s).$$

Therefore

$$V_{xx} = 2e^{2r(T-t)}, \quad V_{sx} = -2e^{2r(T-t)}\Delta^{BS}(t, s).$$

The deterministic HJB feedback is

$$\Delta^{HJB}(t, s) = \frac{\sigma^2 s^2 e^{2r(T-t)}}{\lambda + \sigma^2 s^2 e^{2r(T-t)}} \Delta^{BS}(t, s).$$

The corresponding entropy-regularized Gaussian control distribution is

$$\Delta^{ER}(t, s) \sim \mathcal{N}\left(\Delta^{HJB}(t, s), \frac{\eta}{2(\lambda + \sigma^2 s^2 e^{2r(T-t)})}\right).$$

Proof. The terminal condition gives $a(T, s) = 1$, $b(T, s) = -2H(s)$, and $c(T, s) = H(s)^2$. Under $\mu = r$, the coefficient equations associated with the quadratic ansatz coincide with the Black-Scholes pricing equation. Hence $a(t, s) = e^{2r(T-t)}$ and $b(t, s) = -2e^{2r(T-t)}C^{BS}(t, s)$. It follows that $V_{xx} = 2e^{2r(T-t)}$ and $V_{sx} = b_s(t, s) = -2e^{2r(T-t)}\Delta^{BS}(t, s)$. Substituting these derivatives into the deterministic feedback formula in Section III gives the HJB hedge. Substituting the same derivatives into (21) gives the Gaussian control distribution. $\square$

B. Connection with deterministic HJB

Entropy regularization keeps the structure of the deterministic HJB problem but changes the form of the control. The controller chooses a distribution whose mean is the deterministic HJB feedback and whose variance is governed by η . In the terminology of Wang, Zariphopoulou, and Zhou [9], the Gaussian mean is the exploitation component, and the variance is the exploration component. This interpretation is useful in the numerical implementation: one can either use the analytical mean feedback or sample a position from the optimal distribution. In the limiting case $\eta \rightarrow 0$, the entropy term vanishes and the Gaussian distribution collapses to a Dirac mass at the deterministic HJB feedback.

C. Verification and admissibility

The derivation above should be read as a formal dynamic-programming characterization. A rigorous verification argument requires sufficient smoothness of the value function, at least $C^{1,2,2}$ in (t, s, x) , so that Ito's formula applies to the coupled state process (S_t, X_t) . The feedback control, or the Gaussian control distribution in the entropy-regularized case, must be admissible: it should generate a unique controlled wealth process, satisfy the needed integrability conditions, and make the objective finite. The terminal condition $V(T, s, x) = (x - H(s))^2$ must hold, and the candidate feedback must attain the local minimum in the HJB equation. Under these regularity and admissibility conditions, the candidate value function can be interpreted as the optimal value, and the Gaussian feedback distribution can be interpreted as the optimal relaxed control. The paper uses this standard verification logic to justify the candidate control; a full PDE verification theorem is left for later work.

V. UTILITY INDIFFERENCE PRICING

A. Background

In the classical Black-Scholes framework, the market is complete and every contingent claim can be perfectly replicated by a self-financing portfolio. The unique no-arbitrage price of a European call option is therefore the Black-Scholes price. With trading frictions and entropy regularization, the market becomes incomplete: perfect replication fails, and residual hedging risk remains at maturity.

In incomplete markets, a common pricing method with a clear theoretical basis is utility indifference pricing. It was introduced by Hodges and Neuberger [7] and later developed in transaction-cost models such as Davis, Panas, and Zariphopoulou [3]. The idea is to find a price that makes the investor indifferent, under the chosen criterion, between holding and not holding the derivative position.

This section uses the entropy-regularized stochastic-control framework from Section IV to interpret the option price. Based on the value function $V(t, s, x)$ in the entropy-regularized HJB equation, I derive the indifference equation and a marginal price approximation. I also show that, when the position penalty λ and temperature η vanish, the indifference price is consistent with the Black-Scholes price.

B. Basic setting

Consider an option seller who decides at $t = 0$ whether to sell a European call option with strike K and maturity T . The terminal payoff is

$$H(S_T) = \max(S_T - K, 0). \quad (22)$$

The seller compares two cases.

1) *Case A: no option position:* The investor has initial wealth x and does not trade the option. Under the risk-neutral measure $\mu = r$, and with a positive position penalty $\lambda > 0$, the optimal benchmark strategy is to hold no risky position: $\delta_t \equiv 0$. Wealth then evolves deterministically as

$$X_T^{(A)} = x \cdot e^{rT}. \quad (23)$$

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Initial underlying price S_0	100
Strike K	100
Risk-free rate r	0.03
Volatility σ	0.20
Maturity T	1 year
Number of rebalancing dates	252
Monte Carlo paths N_{MC}	100,000
Random seed	20260529
Baseline position penalty λ	50
Baseline entropy temperature η	0.05

corresponds to an average exploration standard deviation of roughly 0.0074 in hedge-ratio units. This is large enough to show the effect of randomized exploration, while still leaving the strategy driven by the analytical feedback.

The numerical implementation uses the analytical controls derived in Sections III and IV, especially Proposition IV.1. The Black-Scholes benchmark uses

$$\Delta^{BS}(t, s) = N(d_1).$$

The deterministic HJB strategy uses the closed-form feedback

$$\Delta^{HJB}(t, s) = \frac{\sigma^2 s^2 e^{2r(T-t)}}{\lambda + \sigma^2 s^2 e^{2r(T-t)}} \Delta^{BS}(t, s). \quad (54)$$

The entropy-regularized strategy uses the Gaussian control distribution

$$\Delta^{ER}(t, s) \sim \mathcal{N}\left(\Delta^{HJB}(t, s), \frac{\eta}{2(\lambda + \sigma^2 s^2 e^{2r(T-t)})}\right). \quad (55)$$

To move from continuous-time feedback to daily trading, let $0 = t_0 < t_1 < \dots < t_N = T$. At each rebalancing time t_n , the program observes S_{t_n} and sets

$$\hat{\Delta}_{t_n}^{HJB} = \frac{\sigma^2 S_{t_n}^2 e^{2r(T-t_n)}}{\lambda + \sigma^2 S_{t_n}^2 e^{2r(T-t_n)}} \Delta^{BS}(t_n, S_{t_n}).$$

For the entropy-regularized strategy, the sampled position is

$$\hat{\Delta}_{t_n}^{ER} = \hat{\Delta}_{t_n}^{HJB} + \sqrt{\frac{\eta}{2(\lambda + \sigma^2 S_{t_n}^2 e^{2r(T-t_n)})}} Z_n^\Delta, \\ Z_n^\Delta \sim N(0, 1).$$

The quantity $\hat{\Delta}_{t_n}$ is the number of shares held on $[t_n, t_{n+1})$. The rebalancing trade is

$$q_n = \hat{\Delta}_{t_n} - \hat{\Delta}_{t_{n-1}}, \quad n \geq 1,$$

with $q_0 = \hat{\Delta}_{t_0}$ as the initial position. Thus daily trading is a time discretization of the analytical feedback. For each strategy, wealth is updated path by path as

$$X_{t_{n+1}} = X_{t_n} + r(X_{t_n} - \Delta_{t_n} S_{t_n}) \Delta t + \Delta_{t_n} (S_{t_{n+1}} - S_{t_n}).$$

The Python script `numerical_experiments.py` generates the tables and figures.

TABLE II
CONTROL-STRUCTURE DIAGNOSTICS IN THE BASELINE CASE

Metric	Value
Average shrinkage factor	0.8898
Average exploration variance	5.51×10^{-5}
Average exploration standard deviation	0.0074

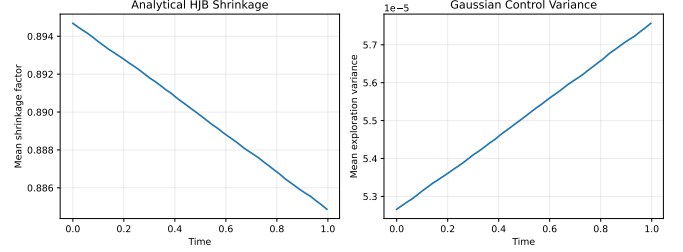


Fig. 1. Time variation of the average shrinkage factor and exploration variance.

B. Structure of the HJB hedge

Equation (54) shows that the HJB hedge is a state-dependent shrinkage of the Black-Scholes delta:

$$\Delta^{HJB}(t, s) = \kappa(t, s; \lambda) \Delta^{BS}(t, s), \\ \kappa(t, s; \lambda) = \frac{\sigma^2 s^2 e^{2r(T-t)}}{\lambda + \sigma^2 s^2 e^{2r(T-t)}}.$$

The shrinkage factor satisfies $0 < \kappa \leq 1$. As $\lambda \rightarrow 0$, $\kappa \rightarrow 1$, and the deterministic HJB strategy converges to the Black-Scholes delta hedge. As λ increases, the stock position is reduced relative to the classical delta. Because the denominator contains $\sigma^2 s^2 e^{2r(T-t)}$, the numerical grid for λ must be interpreted relative to the scale of $\sigma^2 S_0^2$. With $S_0 = 100$ and $\sigma = 0.20$, this scale is around 400, so the sensitivity experiment includes λ values of comparable size.

Table II gives diagnostics tied directly to the analytical formula. In the baseline case $\lambda = 50$ and $\eta = 0.05$, the average HJB shrinkage factor is 0.8898. The deterministic HJB hedge therefore holds about 89% of the Black-Scholes delta on average. The average exploration variance is 5.51×10^{-5} , corresponding to an average exploration standard deviation of 0.0074 in hedge-ratio units. The baseline entropy-regularized strategy is close to the deterministic HJB mean, but still contains randomized exploration.

Figure 1 shows the evolution of these quantities on the rebalancing grid. The left panel tracks the average shrinkage factor. The right panel tracks the average variance of the Gaussian control distribution. The two panels separate the channels of the model: λ reduces the central hedge ratio, and η controls randomized exploration around that central hedge.

C. Entropy-regularized control distribution

The entropy-regularized strategy in (55) has two parts. Its mean is the deterministic HJB feedback $\Delta^{HJB}(t, s)$, the

TABLE III
TERMINAL P&L AND PENALIZED OBJECTIVE FOR THE THREE HEDGING STRATEGIES

Strategy	Mean P&L	Std. P&L	MSE	VaR 95%	CTE 95%	VaR 99%	CTE 99%	$E \int \Delta^2 dt$	$\lambda E \int \Delta^2 dt$	J_λ
Black-Scholes Delta	0.0014	0.4391	0.1928	0.7127	1.0016	1.1726	1.4612	0.4036	20.1812	20.3740
Deterministic HJB	-0.0031	1.3675	1.8700	2.7250	3.3847	3.8023	4.2825	0.3339	16.6925	18.5625
Entropy-regularized HJB	-0.0037	1.3754	1.8916	2.7354	3.3973	3.8283	4.3024	0.3339	16.6953	18.5869

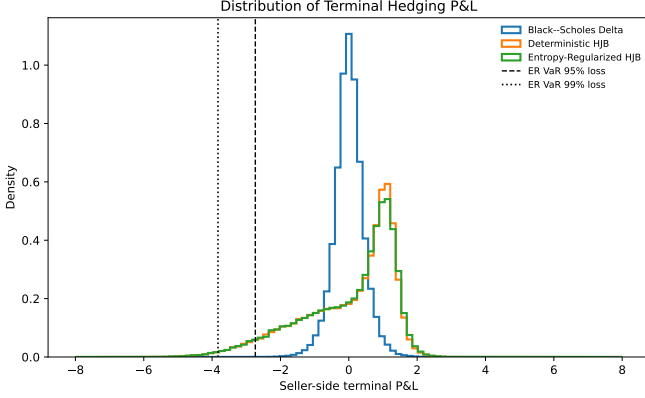


Fig. 2. Distribution of seller-side terminal hedging P&L.

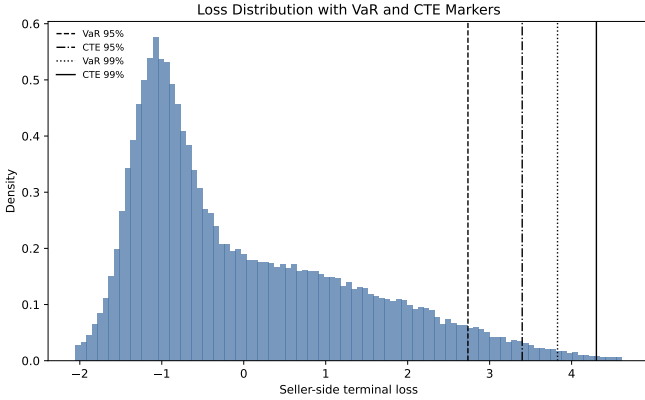


Fig. 3. Terminal loss distribution under the entropy-regularized strategy.

G. Downside risk

Figure 3 focuses on the terminal loss distribution $L_T = -\Pi_T^{\text{seller}}$ for the entropy-regularized strategy. The VaR markers identify quantile losses, while the CTE markers show the average loss beyond the corresponding VaR threshold. This distinction matters because two strategies may have similar VaR but different average losses past the quantile. In this experiment, the 99% VaR of the entropy-regularized strategy is approximately 3.8283, and the 99% CTE is approximately 4.3024, so the extreme tail is larger than the quantile alone suggests.

H. Parameter sensitivity

Table IV reports sensitivity to the position penalty. This experiment is included because a single baseline value $\lambda = 50$

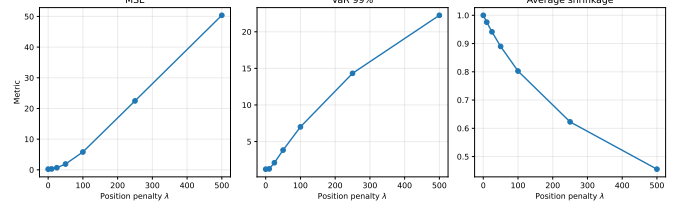


Fig. 4. Risk metrics as the position penalty changes.

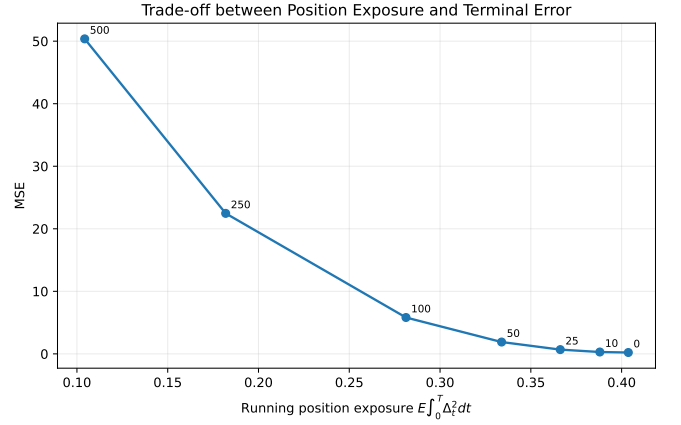


Fig. 5. Tradeoff between position exposure and terminal MSE.

does not explain what the parameter controls. As λ increases from 0 to 500, the average shrinkage factor falls from 1.0000 to 0.4554, and the running position exposure $E \int \Delta^2 dt$ falls from 0.4037 to 0.1043. This directly confirms the interpretation of (54): the HJB feedback is a penalized delta hedge in which λ reduces the central position. The cost is larger terminal error. MSE rises from 0.2177 at $\lambda = 0$ to 50.3690 at $\lambda = 500$, and the 99% CTE rises from 1.5049 to 25.7098. The values of J_λ in the table should be read as objective values under different penalty parameters. In this Black-Scholes benchmark, increasing λ lowers exposure but worsens replication quality.

Figure 4 shows the same pattern. As λ increases, the average shrinkage factor decreases, while MSE and tail risk increase. This agrees with the analytical formula for $\kappa(t, s; \lambda)$.

Economically, this is a Pareto tradeoff. A larger λ lowers risky stock exposure, but it also moves the strategy away from frictionless replication. Figure 5 plots this tradeoff directly. Moving along the curve from $\lambda = 0$ to larger values lowers $E \int_0^T \Delta_t^2 dt$, but the lower exposure is bought with higher

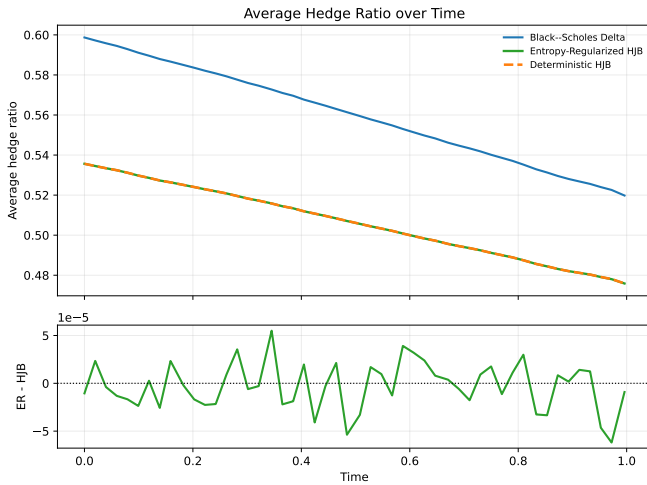


Fig. 9. Average hedge ratio under the three strategies.

K. Interpretation

The main lesson from the numerical experiment is that entropy regularization must be understood relative to the objective function. In a correctly specified Black-Scholes market without transaction costs, the classical delta hedge remains the most accurate strategy if the criterion is only terminal replication error. The numerical results are consistent with this. The model here is better understood as a penalized extension of the Black-Scholes delta.

The deterministic HJB feedback is a penalized delta hedge. Through the shrinkage factor $\kappa(t, s; \lambda)$, it deliberately reduces the stock position. This increases terminal MSE but lowers running position exposure, and in the baseline case $\lambda = 50$, it lowers the reported penalized objective J_λ . The entropy-regularized strategy then adds a state-dependent Gaussian exploration variance around this analytical mean. In the present experiment, exploration slightly widens the P&L distribution. Its value lies in placing position shrinkage, randomized hedge dispersion, terminal P&L, VaR, CTE, the original penalized objective, and approximate indifference bid-ask spreads in one interpretable stochastic-control framework.

VII. CONCLUSION

This paper studies option hedging with market frictions from a stochastic-control perspective. Using the Black-Scholes delta hedge as a benchmark [1], [2], I formulate a deterministic HJB hedging model with a quadratic position penalty to describe the tradeoff between replication accuracy and trading intensity [3]–[6]. I then introduce entropy regularization, relaxing the deterministic position into a probability distribution over admissible actions [9].

The main theoretical result is the derivation of the entropy-regularized HJB equation and the closed-form characterization of the optimal exploratory control. On an unconstrained action space, the optimal distribution is Gaussian. Its mean equals the deterministic HJB feedback control, and its variance is gov-

erned by the temperature parameter. This gives a direct link between deterministic optimal hedging and entropy-regularized randomized control, and it shows that the model reduces to the deterministic HJB formulation as the temperature tends to zero.

The numerical results confirm the economic interpretation of the model and clarify its limitations. In the correctly specified Black-Scholes benchmark, the classical delta hedge produces the smallest P&L dispersion, MSE, VaR, and CTE. The analytical deterministic HJB feedback reduces hedge aggressiveness through a state-dependent shrinkage factor and lowers running position exposure, but it increases terminal hedging risk. The entropy-regularized model adds Gaussian exploration around this deterministic mean. In the reported experiments, increasing the temperature widens the P&L distribution and increases tail losses. The Monte Carlo indifference-pricing experiment further shows how residual hedging risk can be converted into ask, bid, and spread adjustments around the Black-Scholes price. The contribution of entropy regularization in this paper is to provide an interpretable stochastic-control tool connecting HJB shrinkage, randomized hedge dispersion, the penalized objective, approximate indifference pricing, and measurable risk quantities such as MSE, VaR, and CTE.

There are several natural extensions. One could add more realistic trading frictions, such as proportional transaction costs or nonlinear market impact. The framework could also be extended to stochastic volatility, jumps, or incomplete-market settings. Combining entropy-regularized control with deep hedging or modern PDE solvers may also improve its use in high-dimensional and path-dependent hedging problems [9]–[11].

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